| | | |
|---|---|---|
| 11(a) | electrons (in gas atoms/molecules) interact with photons | **B1** |
| | photon energy causes electron to move to higher energy level/to be excited | **B1** |
| | photon energy = difference in energy of (electron) energy levels | **B1** |
| | when electrons de-excite, photons emitted in all directions (so dark line) | **B1** |
| 11(b)(i) | photon energy $\propto 1 / \lambda$ | **C1** |
| | energy = 1.68 eV | **A1** |
| | **or** | |
| | $E = hc / \lambda$<br>$E = 6.63 \times 10^{-34} \times 3.0 \times 10^{8} / (740 \times 10^{-9})$<br>$= 2.688 \times 10^{-19}$ J | **(C1)** |
| | energy = 1.68 eV | **(A1)** |
| 11(b)(ii) | 3.4 eV $\rightarrow$ 1.5 eV<br>3.4 eV $\rightarrow$ 0.85 eV<br>3.4 eV $\rightarrow$ 0.54 eV<br>*all correct and none incorrect 2/2*<br>*2 correct and 1 incorrect or only 2 correctly drawn 1/2* | **B2** |